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Downhole setting tool with exhaust diffuser

Abstract

A method for redressing a setting tool for actuating a plug in a subterranean wellbore includes recovering the setting tool from the wellbore, the setting tool including a housing with a throughbore therein, a piston arranged for axial movement within the throughbore, a combustible element, and an exhaust diffuser configured to redirect a flow of combustion products generated in response to ignition of the combustible element, removing the piston from the throughbore of the housing, removing the exhaust diffuser from the housing, cleaning the piston and the housing, and reinstalling the piston into the housing of the setting tool with at least one of the exhaust diffuser and a replacement exhaust diffuser for subsequent use of the setting tool in a wellbore.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a division of U.S. non-provisional patent application Ser. No. 17/837,880 filed Jun. 10, 2022, entitled “Downhole Setting Tool with Exhaust Diffuser”, which claims benefit of U.S. provisional patent application No. 63/209,195 filed Jun. 10, 2021, entitled “Downhole Setting Tool with Exhaust Diffuser,” both of which are hereby incorporated herein by reference in their entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT
(1) Not applicable.
BACKGROUND
(2) During completion operations for a subterranean wellbore, it is conventional practice to perforate the wellbore with perforating guns along with any casing tubulars disposed therein along a targeted hydrocarbon bearing formation to provide a path for formation fluids (e.g., hydrocarbons) to flow into the wellbore. To enhance the productivity of each of typically a great many perforations, the wellbore is divided into a plurality of production zones along the targeted formation where the perforations associated with each zone are enlarged and expanded by hydraulic fracturing sometimes referred to as “fracking”. Each production zone is isolated from the other downhole zones using a sealing device (e.g., a plug, a packer) installed within the wellbore prior to the given production zone being perforated.

(3) Generally, both a setting tool and at least one perforating gun assembled along the same tool string are inserted into the wellbore in order to set the sealing device and then perforate the casing in a single trip downhole. The setting tool typically includes an explosive or combustible element for shifting the sealing device within the wellbore from an initial configuration in which fluid flow is permitted around the sealing device and a set configuration in which the sealing device plugs the wellbore. With the sealing device in the set configuration the setting tool separates from the set sealing device to permit the setting tool to be pulled back to the surface with the rest of the tool string.
(4) In the process of setting the sealing device, the combustible element is typically used to power the setting tool and thereby drive the operable elements of the sealing device. The combustion process within the setting tool undesirably results in dramatic heating of the setting tool and wellbore fluids within the vicinity. Particularly, the combustion process generates high-velocity combustion products which may, if unimpeded, peel protective coatings or otherwise damage surfaces of the setting tool including surfaces relied on for sealing different internal chambers of the setting tool. Additionally, the dramatic heating of the setting tool and local wellbore fluids stimulates reactions between constituents of the combustion products and the heated wellbore fluids which may result in the undesirable deposition of hard reaction products which may tightly adhere onto surfaces of the setting tool. Typically, the mineral deposits formed on the setting tool must be cleaned off after use of the setting tool and before any subsequent use of the setting tool. Given the hardness of these mineral deposits and how tightly they adhere to the setting tool, removing of the mineral deposits can be a difficult, costly, and time-consuming process. For at least these reasons, any reformulation of the wellbore fluids, combustion materials or setting tools that might reduce the formation of such hard and challenging mineral deposits onto the setting tool would be greatly appreciated in the industry.
SUMMARY OF THE DISCLOSURE
(5) An embodiment of a method for redressing a setting tool for actuating a plug in a subterranean wellbore comprises (a) recovering the setting tool from the wellbore, the setting tool comprising a housing with a throughbore therein, a piston arranged for axial movement within the throughbore, a combustible element, and an exhaust diffuser configured to redirect a flow of combustion products generated in response to ignition of the combustible element, (b) removing the piston from the throughbore of the housing, (c) removing the exhaust diffuser from the housing, (d) cleaning the piston and the housing, and (e) reinstalling the piston into the housing of the setting tool with at least one of the exhaust diffuser and a replacement exhaust diffuser for subsequent use of the setting tool in a wellbore. In some embodiments, (e) comprises orienting at least a portion of a baffle face of at least one of the exhaust diffuser and the replacement exhaust diffuser at an angle that is between 60° and 120° relative to a longitudinal axis of an exhaust port of the piston. In some embodiments, (e) comprises positioning at least one of the exhaust diffuser and the replacement exhaust diffuser in an annular expansion chamber formed radially between the outer surface of the piston and the inner surface of the housing. In certain embodiments, (e) comprises positioning at least one of the exhaust diffuser and the replacement exhaust diffuser in an internal firing chamber formed within the piston. In certain embodiments, (e) comprises releasably coupling at least one of the exhaust diffuser and the replacement exhaust diffuser to the piston. In some embodiments, (e) comprises releasably coupling at least one of the exhaust diffuser and the replacement exhaust diffuser to the housing. In some embodiments, (d) comprises cleaning the exhaust diffuser, and (e) comprises reinstalling the piston into the housing of the setting tool with the cleaned exhaust diffuser. In certain embodiments, (e) comprises reinstalling the piston into the housing of the setting tool with the replacement exhaust diffuser.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a detailed description of exemplary embodiments of the disclosure, reference will now be made to the accompanying drawings in which:

(2) FIG. **1** is a schematic, partial cross-sectional view of a system for completing a subterranean well including an embodiment of a setting tool;

(3) FIG. **2**A is a partial cross-sectional view of a system for completing a subterranean well including an embodiment of a setting tool;

(4) FIG. **2**B is a schematic view of a system for completing a subterranean well including another embodiment of a setting tool;

(5) FIG. **3** is a side cross-sectional views of an embodiment of a setting tool;

(6) FIG. **4** is a cross-sectional view of an embodiment of an exhaust diffuser of the setting tool of FIG. **3**;

(7) FIGS. **5-7** are side views of the exhaust diffuser of FIG. **4**;

(8) FIGS. **8** and **9** are partial cross-sectional views of the setting tool of FIG. **3** partially stroked;

(9) FIG. **10** is a partial cross-sectional view of the setting tool of FIG. **3** maximally stroked;

(10) FIG. **11** is a partial cross-sectional view of another embodiment of a setting tool;

(11) FIG. **12** is a partial cross-sectional view of another embodiment of a setting tool;

(12) FIG. **13** is a cross-sectional view of another embodiment of a setting tool;

(13) FIG. **14** is a perspective view of a flow plug side another embodiment of a setting tool; and

(14) FIG. **15** is a block diagram of an embodiment of a method for redressing a setting tool for actuating a plug in a subterranean wellbore.

DETAILED DESCRIPTION

(15) The following discussion is directed to various exemplary embodiments. However, one skilled in the art will understand that the examples disclosed herein have broad application, and that the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to suggest that the scope of the disclosure, including the claims, is limited to that embodiment. Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not function. The drawing figures are not necessarily to scale. Certain features and components herein may be shown exaggerated in scale or in somewhat schematic form and some details of conventional elements may not be shown in interest of clarity and conciseness.

(16) In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection, or through an indirect connection via other devices, components, and connections. In addition, as used herein, the terms “axial” and “axially” generally mean along or parallel to a central axis (e.g., central axis of a body or a port), while the terms “radial” and “radially” generally mean perpendicular to the central axis. For instance, an axial distance refers to a distance measured along or parallel to the central axis, and a radial distance means a distance measured perpendicular to the central axis. Any reference to up or down in the description and the claims is made for purposes of clarity, with “up”, “upper”, “upwardly”, “uphole”, or “upstream” meaning toward the surface of the borehole and with “down”, “lower”, “downwardly”, “downhole”, or “downstream” meaning toward the terminal end of the borehole, regardless of the borehole orientation. Further, the term “fluid,” as used herein, is intended to encompass both fluids and gasses.

(17) Tools used in oil well or gas wells are introduced or carried into a subterranean wellbore on a workstring, such as wire line, electric line, continuous coiled tubing, threaded workstring, or the

like, for engagement at a pre-selected position within the wellbore. The wellbore can be lined with a tubular conduit such as a casing string or liner. The wellbore can be an openhole section where the drilled formation does not have the conduit supporting the drilled formation. The wellbore can include a secondary tubing member, such as production tubing, that is placed within a casing, liner, or openhole section. These completion tools include sealing devices such as expandable elastomeric plugs, permanent or retrievable plugs, packers, ball-type and other valves, injectors, perforating guns, tubing and casing hangers, cement plug dropping heads, and other devices typically encountered during the drilling, completion, or remediation of a subterranean well. Such devices and tools will hereafter collectively be referred to as “auxiliary tools.” The auxiliary tool is typically set and anchored into position within the casing, tubing, or openhole section such that movements in various directions such as upwardly, downwardly, or rotationally, are resisted, and, in fact, prevented. Such movements can occur as a result of a number of causes, such as pressure differentials across the tool, temperature variances, tubing or other conduit manipulation subsequent to setting for activation of other tools in the well, and the like.

(18) The auxiliary tool typically must be set or actuated to position the auxiliary tool at the required depth within the casing, liner, tubing, or openhole section. In some cases, the auxiliary tool may comprise, for example, a plug or packer including a packing element that will form a seal when energized. As described above, the activation or manipulation of some of such auxiliary tools often is achieved by use of a setting tool which can be introduced into the wellbore along with or subsequent to the auxiliary tool on a workstring, such as wire or electric line, continuous or coiled tubing, threaded tubing, drill pipe, or by other known means. In some applications, the setting tool includes a piston to move or stroke a portion of the setting tool relative to stationary portion of the setting tool to apply a setting force in compression or in tension to the auxiliary tool. Pressure can be applied to face of the piston within the setting tool to generate the setting force to set or actuate the auxiliary tool.

(19) Some setting tools utilize an explosive or combustible charge or element to develop a high-pressure gas within a firing chamber of the setting tool following ignition of the combustible element. The high pressure generated by the burning or firing of the pyrotechnic charge drives a piston, stroking rod, or other member of the setting tool to move relative a stationary member to cause the manipulation of the auxiliary tool. By “burning” or “firing” it is meant the continuous generation, sometimes relatively slowly, of pressure by ignition of a power charge initiated reaction which results in a pressure increase within a firing chamber of transmittable gaseous pressure within the apparatus. The term “detonate” can also be used to describe a sudden generation of gaseous pressure. Sometimes the terms “detonate” and “ignite” are used to describe a sudden generation of gaseous pressure. The terms “detonate”, “burning”, “igniting,” or “firing”, all describe the generation of gaseous pressure by the burning of the combustible element with different timescales.

(20) The ignition of the combustible element to burn is started with an igniter. The igniter can be comprised of a plurality of igniters. For example, the igniter can be a single primary igniter, a primary igniter and secondary igniter, or a primary, secondary, and embedded igniter. The primary igniter and/or secondary igniter can comprise a tube, an electronic ignition device, and a pyrotechnic material that creates a jet of heat and flame. In some embodiments, the igniter can be installed within a firing head or setting tool initiator and connected to the firing chamber of the setting tool. The upper end of the firing head can couple to any combination of a cable head, an instrument sub, a quick connect sub, a switch sub, or any other type of with a threaded connection and an electrical connection.

(21) In a typical deployment of a toolstring including a conventional setting tool, one or more operators at the surface prepare the toolstring for conveyance into the wellbore. The toolstring can comprise, among other things, a workstring, the conventional setting tool, and an auxiliary tool. The operator may releasably connect the auxiliary tool to the setting tool, install a power charge

into the firing chamber of the setting tool, and connect the setting tool initiator to the conventional setting tool. The operator may then direct the conveyance of the tool sting into the wellbore via the workstring and convey the toolstring to the desired location. The location of the toolstring can be verified by any combination of a measured length of the toolstring and the number of collars counted by a collar locator.

(22) Once at the desired location, the operator may signal an initiator switch of the setting tool initiator to ignite the igniter to activate the conventional setting tool to set the auxiliary tool at the desired location. The combustible element burns in response to ignition of the igniter to produce high pressure and high temperature combustion products that can corrode, erode, or otherwise damage surfaces of the setting tool. In some instances, damage may occur from the combustion products to protective coatings formed on surfaces of the setting tool. The erosion of the protective coating can cause corrosion of sealing surfaces of the setting tool and thereby undesirably shorten the operational lifespan of the conventional setting tool. Moreover, hot combustion products may form mineral deposits upon the surfaces of the setting tool, which may be time consuming or impractical to remove, also shortening the operational life of the conventional setting tool. The elevated heat of the combustion products may cause some of them to bond to surfaces of the setting tool, leading to rapid corrosion. An operator of the toolstring may require the setting of twenty or more auxiliary tools using one or more setting tools in a given application. Damage to the setting tool from the hot and the subsequent shortening of the service life of the setting tool can prevent the operator from providing an efficient and reliable plugging and perforation of the wellbore.

(23) Thus, it is desirable to develop a setting tool configured to minimize or prevent damage which occurs thereto following ignition of the combustible element of the setting tool.

(24) Embodiments described herein include a setting tool comprising an exhaust diffuser positioned along a flowpath of combustion products generated by the detonation of a combustible element of the setting tool. The exhaust diffuser reduces the velocity of the combustion products flowing along the combustion product flowpath to thereby reduce the power which the combustion products contact components of the setting tool, such as seal surfaces of a housing of the setting tool. Reducing the velocity and attendant power the combustion products may project against components of the setting tool reduces the abrasive potential of the combustion products, thereby preventing or at least mitigating damage (e.g., the peeling of protective coatings) that occurs to components of the setting tool in response to contact with the combustion products. Additionally, by mitigating the power of the combustion products before they are permitted to contact at least some surfaces of the setting tool (e.g., seal surfaces of the setting tool), the issue of combustion products bonding to surfaces of the setting tool may also be eliminated or at least substantially mitigated. Further, the exhaust diffuser may act to trap debris resulting from detonation of the combustible element (e.g., material remains of the combustible element) within a combustion or firing chamber of the setting tool, preventing those debris from percolating through the setting tool, making the setting tool significantly easier to clean and redress before being reused in the same or a different wellbore.

(25) Embodiments of setting tool exhaust diffusers described herein include one or more baffle faces and one or more passages positioned downstream from the one or more baffle faces along the combustion product flowpath. The one or more passages to direct the high-pressure and high-temperature combustion products away from protective coatings formed on the seal surfaces of the setting tool. The one or more passages of the exhaust diffuser are configured to diffuse the flow of high-pressure and high-temperature combustion products away from impinging directly onto the seal surface. The passages can be made of corrosion resistant materials and reused. The passages can also be removed and replaced as needed. The setting tool can be cleaned, inspected, and redressed on location (or elsewhere) after usage in the wellbore by service personnel. The passages of the exhaust diffuser protects the piston of the setting tool during usage to increase the life of the assembly and greatly reduce the number of setting tool that fail inspection after usage.

(26) Referring now to FIGS. **1** and **2**, an embodiment of a system **10** for plugging a wellbore **14** extending through a subterranean earthen formation **16** is shown. In this exemplary embodiment, system **10** includes a surface assembly or servicing rig **12** that extends over and around the wellbore **14** that penetrates the earthen formation **16** for the purpose of recovering hydrocarbons from a first production zone **18**A and a second production zone **18**B (collectively the production zones "18"). The wellbore **14** can be drilled into the earthen formation **16** using any suitable drilling technique. While shown as extending vertically from the surface in FIG. **1**, the wellbore **14** can also be deviated, horizontal, and/or curved over at least some portions of the wellbore **14**. For example, the wellbore **14**, or a lateral wellbore drilled off of the wellbore **14**, may deviate and remain within one of the production zones **18**. The wellbore **14** can be cased, open hole, contain tubing, and can generally be made up of a hole in the ground having a variety of shapes and/or geometries as is known to those of skill in the art. In the illustrated embodiment, a casing string **20** made up of multiple sections of threaded pipe joined with threaded couplings can be placed in the wellbore **14** and secured at least in part by cement **22**.
(27) The servicing rig **12** of system **10** can be one of a drilling rig, a completion rig, a workover rig, a wireline system, or other structure and supports a toolstring **32** in the wellbore **14**. Servicing rig **12** includes a surface controller **13** in signal communication with one or more downhole tools of toolstring **32**. In other embodiments, other surface systems or structures can also support the toolstring **32**. The servicing rig **12** can also comprise a derrick with a rig floor through which the toolstring **32** extends downward from the servicing rig **12** into the wellbore **14**. It is understood that other mechanical mechanisms, not shown, can control the run-in and withdrawal of the toolstring **32** in the wellbore **14**.
(28) In this exemplary embodiment, toolstring **32** generally includes a workstring **30**, one or more perforating guns **46** (hidden from view in FIG. **2**), a signal sub **34**, a setting tool **42**, and an auxiliary tool **44**. It may be understood that in other embodiments the configuration of tool string **32** may vary. For example, in some embodiments, tool string **32** may additionally include a fishneck, one or more weight bars, a release tool, and/or one or more other downhole tools. The workstring **30** can be any of a string of jointed pipes, a slickline, a coiled tubing, and a wireline. Auxiliary tool **44** may comprise one or more frac plugs, one or more packers, one or more tubing hangers, one or more completion components such as screens and/or production valves, sensing and/or measuring equipment, and other equipment which are not shown in FIGS. **1** and **2**. The toolstring **32** can be lowered into the wellbore **14** to position the setting tool **42** to set or actuate a frac plug at a predetermined depth.
(29) In this exemplary embodiment, cable head **36** is the uphole-most component of toolstring **32** and includes an electrical connector for providing electrical signal and power communication between the workstring **30** and the other components (e.g., instrument sub **38**, setting tool initiator **40**, setting tool **42**, etc.) of toolstring **32**. The instrument sub **38** can contain one or more environmental sensors **56**. For example, the instrument sub **38** can include a magnetic sensor (e.g., a casing collar locator (CCL)), a temperature sensor, a pressure sensor, or a motion sensor (e.g., an accelerometer). The magnetic sensor, generally referred to as a CCL, is generally configured to transmit an electrical signal to the surface via workstring **30** when the CCL passes through a casing collar, where the transmitted signal may be recorded at the surface as a collar kick to determine the position of toolstring **32** within wellbore **14** by correlating the recorded collar kick with an open hole log.
(30) Additionally, in this exemplary embodiment, a setting tool initiator **40** is coupled to a downhole end of instrument sub **38** and is generally configured to provide a connection between the instrument sub **38** and the setting tool **42**. Setting tool initiator **40** couples the cable head **36** of the toolstring **32**, via the instrument sub **38**, to the setting tool **42** and an auxiliary tool **44**, and is generally configured to pass electronic signals and/or power from the conductor **28** within the workstring **30** to an igniter within the toolstring **32**. Setting tool initiator **40** may also include

mechanical and/or electrical components to fire the setting tool **42**.

(31) In this exemplary embodiment, setting tool **42** is coupled to a downhole end of setting tool initiator **40** and is generally configured to set or install auxiliary tool **44** within casing string **20** to isolate desired segments of the wellbore **14**, as will be discussed further herein. Typically, the auxiliary tool **44** is intended to be set or actuated to position the auxiliary tool **44** at the required depth within the wellbore **14**. The actuation or setting of the auxiliary tool **44** may displace a portion of the auxiliary tool relative to another portion to anchor or position the auxiliary tool **44** to a location within the wellbore.

(32) In some embodiments, auxiliary tool **44** comprises a slip that engages or grips the casing string **20**. In some embodiments, auxiliary tool **44** also includes a packing element, generally formed of an elastomeric material configured to seal against the casing string **20** when compressed or energized. Particularly, the packing element may form a seal against the inner surface of casing string **20** to restrict fluid communication through wellbore **14** across the auxiliary tool **44**. The auxiliary tool **44** may be any suitable downhole tool or frac plug known in the art while still complying with the principles disclosed herein. Additionally, it may be understood that although setting tool **42** is shown in FIGS. **1** and **2** as incorporated in toolstring **32**, setting tool **42** may be used in other toolstrings which vary in configuration from toolstring **32**. Following the setting of the auxiliary tool **44**, shaped explosive charges of the one or more perforating guns **46** may be detonated at a desired location in the wellbore **14**.

(33) Turning now to FIG. **3**, an embodiment of a setting tool **100** is shown. In some embodiments, setting tool **42** shown in FIGS. **1** and **2** may be configured similarly as the setting tool **100** shown in FIG. **3**. Additionally, setting tool **100** may be utilized in toolstrings which vary from the configuration of toolstring **32** shown in FIGS. **1** and **2**. In this exemplary embodiment, setting tool **100** has a central or longitudinal axis **105** and generally includes an upper connector **106**, a piston **104**, and a housing **108** in which the piston **104** is slidably positioned. Piston **104** is generally cylindrical having an outer surface **110**, an internal bore or passage **112**, a first or uphole end **114**, and a second or downhole end **116** opposite the uphole end **114**. The internal bore **112** of piston **104** extends centrally through piston **104** and is defined by a generally cylindrical inner surface **118** which extends a portion of the length from the uphole end **114** to the downhole end **116**. The piston **104** has an annular seal assembly **122** comprising a pair of annular seals disposed on an outer surface **120**. The piston **104** further includes an outer surface **124** located proximal to the downhole end **116**. The outer surface **124** has a first recess **126** proximate the annular seal assembly **122** and a second recess **128** proximate the downhole end **116**.

(34) The upper connector **106** of setting tool **100** has a generally cylindrical shape with an outer surface **130** and an inner surface **132**. The upper connector **106** is releasably coupled to the piston **104** via a threaded connection **134**. The threaded connection **134** includes an internal connector **136** of the upper connector **106**, an external connector **138** of the piston **104**, and annular seals **140** in sealing engagement with piston **104**. In an embodiment, upper connector **106** can be combined with piston **104** to form a unitary body. Additionally, it may be understood that the configuration of upper connector **106** may vary in other embodiments.

(35) The housing **108** of setting tool **100** is generally cylindrical and slidably disposed on the piston **104** and has an outer surface **144** and an inner surface **146** defining a throughbore of the housing **108**. In this exemplary embodiment, housing **108** comprises a first or uphole section housing **147** and a second or downhole housing section **148** coupled end-to-end to the uphole section housing **147** by a threaded connection **150** to form the housing **108** whereby relative axial movement is restricted between housing sections **147** and **148**. It may be understood that in other embodiments housing **108** may comprise a single, integrally or monolithically formed housing while in other embodiments housing **108** may comprise more than two separate housing sections connected end-to-end in a manner similar to the connection formed between section housings **147** and **148**.

(36) Downhole housing section **148** of housing **108** interfaces with the auxiliary tool **44** and thus

may also be referred to herein as housing adapter **148**. Threaded connection **150** can include an external connection **152** of the housing adapter **148**, an internal connection **154** of the uphole housing section **147**, and an annular seal assembly **156** in sealing engagement with the uphole housing section **147**. The housing adapter **148** of housing **108** can include a seal assembly **158** and a wiper seal **160** in sealing engagement with outer surface **124** of the piston **104**. Generally, housing **108** includes a seal assembly **164** in sealing engagement with the outer surface **110** of the piston **104**. In some embodiments, housing **108** may be secured to the upper connector **106** by a shear screw **168** to initially restrict relative axial movement between housing **108** and piston **104**. The shear screw **168** may be threadingly connected to the housing **108** and disposed into a recess **170** on the upper connector **106**. The shear screw **168** can be a frangible connector that shears, e.g., breaks, at a predetermined shear stress amount. The shear screw **168** can retain the housing **108** in a position relative to the upper connector **106** until an axial force of a predetermined amount shears or breaks to the shear screw **168** as will be described further herein.

(37) In this exemplary embodiment, housing **108** of setting tool **100** defines an internal balance chamber **174**, an internal firing chamber **176**, and an internal expansion chamber **178**. The balance chamber **174** comprises an annular space defined by the inner surface **146** of the housing **108**, the outer surface **110** of the piston **104**, the seal assembly **164** of the housing **108**, and the annular seal assembly **122** of the piston **104**. At least a portion of the inner surface **146** comprises a seal surface against which seal assembly **122** sealingly engages. Additionally, in some embodiments, at least a portion of the inner surface **146** may be defined by a protective coating intended to protect the inner surface **146** from damage (e.g., corrosion) during the operation of setting tool **100**. The balance chamber can initially contain air at atmospheric or near atmospheric pressure.

(38) The firing chamber **176** of the setting tool **100** is located in the internal bore **112** of the piston **104**. In some embodiments, the firing chamber **176** also includes at least a portion of the inner bore **172** of the upper connector **106**. The firing chamber **176** can contain air at atmospheric or near atmospheric pressure and receives a combustible element **180** (shown only schematically in FIG. **3**) disposed therein. Combustible element **180** may comprise a pyrotechnic or “black powder” charge that comprises a mixture of gun powder or black powder, filler material, and an oxidizer, similar to a road flare, that is slidingly fits into the firing chamber. Combustible element **180** is configured to produce high-pressure and high-temperature combustion products within the firing chamber **176** in response to being ignited by an igniter (e.g., an igniter of setting tool initiator **40**). A high-pressure gas can be generated by the burning or firing of the combustible element **180**. By “burning” or “firing” it is meant the continuous generation, sometimes relatively slowly, of gas pressure by ignition of a power charge which results in a pressure increase within a firing chamber of transmittable gaseous pressure within the apparatus. Sometimes the term “detonate” is used to describe a sudden generation of gaseous pressure. The terms “detonate”, “burning”, or “firing”, all describe the generation of gaseous pressure by the burning of the power charge with different timescales.

(39) The expansion chamber **178** of the setting tool **100** includes an annular space defined by the inner surface **118** of the housing **108**, the first recess **126** of the piston **104**, the annular seal assembly **122** of the piston **104**, and the seal assembly **158** of the housing adapter **148**. The expansion chamber **178** may initially contain air at atmospheric or near atmospheric pressure. In this exemplary embodiment, piston **104** comprises one or more internal exhaust ports **182** which extend at an incline from the firing chamber **176** to the expansion chamber **178**, thereby fluidically connecting the firing chamber **176** to the expansion chamber **178**. Particularly, the exhaust ports **182** of piston **104** extend at an acute angle to a central axis of the setting tool **100** from the inner surface **118** of inner bore **112** to an annular shoulder formed on the outer surface **110** of piston **104**. While in this exemplary embodiment the exhaust ports **182** of piston **104** are inclined relative to a central axis of the piston **104**, in other embodiments, exhaust ports **182** may extend only axially, only radially, or in one or more different directions. In embodiments in which exhaust ports **182**

extend radially through piston **104**, a baffle face **200** of an exhaust diffuser **184** (will be described further herein) may be oriented in the direction of the central axis of piston **104**, orthogonal the radial flow of combustion products exiting the radially extending exhaust ports **182**.

(40) During operation of the setting tool **100**, the combustible element **180** is ignited by an igniter. The ignition of the combustible element **180** produces high-pressure and high-temperature combustion products within the firing chamber **176**. The combustion products flow through exhaust ports **182** and into expansion chamber **178** as pressure builds within the firing chamber **176** of the piston **104**. The pressure within the expansion chamber **178**, created by the burning of the combustible element **180**, acts against an axially-projected piston area of the uphole end the housing adapter **148**, defined by the inner surface **145** of the housing **108** and the outer surface **124** of the piston **104**, to move the housing **108** relative to the piston **104**. The combustion products exiting the exhaust ports **182** can impinge on the inner surface **146** of the housing **108** and may damage or degrade a portion of the inner surface **146**, including portions of inner surface **146** defined and protected by a protective coating applied. Exposure of the uncoated inner surface **146** of the housing **108** to wellbore fluids can induce corrosion and shorten the service life of the housing **108**. Additionally, combustion products may form mineral deposits on setting tool **100** including on the inner surface **146**. Damage to inner surface **146** may prevent seal assembly **122** from forming a seal against one or more portions of the inner surface **146**, permitting combustion products to undesirably leak across the interface formed between seal assembly **122** and inner surface **146**, thereby lowering the setting force which may be generated by setting tool **100** for setting the auxiliary tool **44**.

(41) The service life of the inner surface **146** of the housing **108** can be extended and maximized by protecting surfaces of setting tool **100**, including inner surface **146**, from impingement with high-pressure and high-velocity combustion products exiting exhaust ports **182**. Turning now to FIGS. **4-7**, it may be initially understood that FIG. **5** is a cross-sectional view of a portion of setting tool **100**, while FIG. **6** is the same portion of setting tool **100** with the cross-sectional view rotated about 45 degrees to intersect one of the exhaust passages **210**.

(42) In this exemplary embodiment, setting tool **100** includes an exhaust diffuser **184** is disposed in the first recess **126** and coupled to piston **104**. Exhaust diffuser **184** is separate and distinct from both the piston **104** and housing **108**, but may be releasably (e.g., via one or more fasteners, snap connectors) or permanently (e.g., via welding, brazing) coupled to either the piston **104** or the housing **108**. For example, the exhaust diffuser **184** can be retained in the first recess **126** by one or more fasteners **186** installed into one or more fastener ports **188** in the piston in the first recess **126** of the piston **104**. In this exemplary embodiment, exhaust diffuser **184** is generally cylindrical having a radially outer surface **192**, a radially inner surface **194**, a baffle face **200**, and an uphole surface **202** which may comprise a portion of the baffle face **200**. Surfaces **192** and **194** extend between longitudinally opposed uphole and downhole ends **196** and **198**, respectively, of the exhaust diffuser **184**. In this exemplary embodiment, the inner surface **194** of exhaust diffuser **184** is held in contact with the first recess **126** by the fastener **186** installed through an aperture **208**. The uphole end **196** of exhaust diffuser **184** may abut the downhole surface **204** of the piston **104**. Additionally, an uphole surface **202** (a portion of the inner surface **194** of the exhaust diffuser **184**) contacts or is positioned adjacent the outer surface **206** of the piston **104**.

(43) It may be understood that exhaust diffuser **184** may comprise a single integrally or monolithically formed member or a plurality of members coupled together. For example, in this exemplary embodiment, the exhaust diffuser **184** comprises a first exhaust diffuser portion **184**A and a second exhaust diffuser portion **184**B coupled to the piston **104** by a first fastener **186**A and a second fastener **186**B. Although the exhaust diffuser **184** is shown consisting of two parts in FIGS. **4-7**, it may be understood the exhaust diffuser **184** can be formed by 1 any number of parts. In this exemplary embodiment, exhaust diffuser **184** includes one or more exhaust passages **210** that extend entirely from the baffle face **200** to the downhole end **198** of the exhaust diffuser **184**. In

this exemplary embodiment, baffle face **200** is annular and planar. Additionally, baffle face **200** is oriented generally orthogonal to a longitudinal axis of each exhaust port **182**. Particularly, in this exemplary embodiment, at least a portion of the baffle face **200** is oriented at an angle of approximately 60° to 120° to the longitudinal axis of each exhaust port **182**; however, the orientation of baffle face **200** relative to exhaust ports **182** may vary in other embodiments. In this arrangement, combustion products impinging on the baffle face **200** are forced to make a substantial change in direction, thereby reducing the velocity of the combustion products in response to impinging against the baffle face **200**. Additionally, exhaust diffuser **184** may trap at least some debris generated by the ignition of combustible element **180** in the firing chamber **176** such that the debris are prevented from entering expansion chamber **178**, making the cleaning and redressing of setting tool **100** substantially easier.

(44) In this exemplary embodiment, exhaust passage **210** comprises a slot that extends radially into the exhaust diffuser **184** from the inner surface **194** to a passage surface **212**. In this configuration, the exhaust passage is defined by passage surface **212**, a first side **214**, and a second side **216** distal from the first side **214**. The passage surface **212** may be curved with a radius about the longitudinal axis with the first side **214** at an angle from the second side **216**. As one example, exhaust passage **210** can be formed by a 15° angle measured from the first side **214** to the second side **216**. It is understood that the 15° angle is an example and the angle can be equal to zero or to a non-zero angle that varies from 15°. Additionally, the first side **214** and second side **216** of the exhaust passage **210** can be flat, curved, or any combination thereof. To provide a few examples, the first side **214** and second side **216** can be formed by a flat surface with an acute angle, co-planar, or obtuse angle measured from a plane that extends from the longitudinal axis of the exhaust diffuser **184**. Additionally, it may be understood that exhaust diffuser **184** can include any number of exhaust passages **210** including a single exhaust passage **210** or zero exhaust passages **210**.

(45) Generally, exhaust diffuser **184** redirects the flow of combustion products from an angular or inclined direction to an axial direction. Turning now to FIGS. **8** and **9**, the passage of combustion products from the firing chamber **176** to the expansion chamber **178** is illustrated by inclined flowpaths **216** extending through exhaust ports **182**, and axial flowpaths **218** extending through exhaust passage **210**. passage

(46) The setting tool **100** is illustrated about mid-stroke in FIGS. **8** and **9** with housing adapter **148** having traveled axially about midway along the outer surface **124** of the piston **104**. In this configuration, the pressure within expansion chamber **178** acts on the cross-sectional area of the housing adapter **148** to move the housing **102** and housing adapter **148** in a first or downhole axial direction **222** relative to the piston **104**. The expansion chamber **178** increases in volume as the housing adapter **148** moves axially in the downhole axial direction **222** relative to the piston **104**. The balance chamber **174** (shown in FIG. **3**) correspondingly decreases in volume as the expansion chamber **178** increases in volume.

(47) In FIG. **8**, combustion products from the burning of the combustible element **180** within the firing chamber **176** pass along inclined flowpaths **216** and through the plurality of exhaust ports **182** to impinge on the baffle face **200** of the exhaust diffuser **184**. In FIG. **9**, the combustion products pass along axial flowpaths **218** and through the plurality of exhaust passages **210** exits to the expansion chamber **178**. The baffle face **200**, oriented generally orthogonal to the exhaust ports **182**, force the combustion products exiting exhaust ports **182** to make a substantial change of direction, thereby reducing the velocity of the combustion products before the combustion products are permitted to contact the surrounding housing **102**.

(48) Additionally, mineral deposits which would otherwise form or collect on surfaces of the housing **102** and piston **104** instead form and collect on surfaces of exhaust diffuser **184**, including the baffle face **200**. In some embodiments, exhaust diffuser **184** may comprise a sacrificial element which may be periodically replaced (suffering from erosion and corrosion due to exposure from the combustion products) while the more expensive and difficult to manufacture housing **102** and

piston **104** are repeatedly reused. In some embodiments, exhaust diffuser **184** may comprise a material which varies from the material from which the housing **102** and/or piston **104** are comprised. For example, housing **102** and/or piston **104** may comprise erosion and/or corrosion resistant materials intended to maximize the operational service life of the housing **102** and/or piston **104**. Conversely, exhaust diffuser **184** may be formed from an inexpensive material not intended to survive exposure to the combustion products for more than a limited number of uses. In this manner the cost of operating setting tool **100** may be minimized by maximizing the operational service life of the housing **102** and piston **104** in exchange for focusing the erosion and corrosion generated by the combustion products onto the sacrificial exhaust diffuser **184** which may be quickly and inexpensively replaced between different uses of the setting tool **100**.

(49) In this manner, the exhaust diffuser **184** redirects the flow of combustion products from the inclined direction along inclined flowpaths **216** to an axial direction along axial flowpaths **218** within the exhaust diffuser **184** defined by the surfaces of the first recess **126**, the passage surface **212**, the first side **214**, and the second side **216**. In this manner, each exhaust passage **210** intersects the baffle face **200** of the exhaust diffuser **184** (shown best in FIG. **7**) to provide a pathway for the flow of combustion products to transition from the inclined flowpath **216** extending through the exhaust ports **182** to the axial flowpath **218** extending through exhaust passages **210**.

(50) Turning now to FIG. **10**, the expansion chamber **178** of setting tool **100** continues to increase in volume as the housing adapter **148** continues to travel in the downhole axial direction **222** until the setting tool **100** reaches maximum stroke as shown particularly in FIG. **10**. The setting tool **100** continues to stroke, e.g., the volume of the expansion chamber **178** continues to increase, until the seal assembly **158** on the housing adapter **148** moves past and out of sealing engagement with the outer surface **124** of the piston **104**.

(51) Turning now to FIG. **11**, another embodiment of an exhaust diffuser **230** is shown. Exhaust diffuser **230** includes features in common with the exhaust diffuser **182** shown in FIGS. **4-7**, and shared features are labeled similarly. In this exemplary embodiment, exhaust diffuser **230** has a cylindrical shape with a radially outer surface **232**, a radially inner surface **234**, a baffle surface **236**, and a downhole surface **238**. Exhaust diffuser **230** has a generally L-shape with a front surface **240** and a top surface **242**. One or more fasteners can attach the exhaust diffuser **230** to a fastener port **188** on the piston **104**.

(52) Turning now to FIG. **12**, another embodiment of an exhaust diffuser **250** is shown. Exhaust diffuser **250** includes features in common with the exhaust diffuser **182** shown in FIGS. **4-7**, and shared features are labeled similarly. In this exemplary embodiment, exhaust diffuser **250** has a ring shape with a radially outer surface **252**, a radially inner surface **254**, an uphole surface **256**, and a downhole surface **258**. The downhole surface **258** of exhaust diffuser **250** abuts the uphole surface **260** of the housing adapter **148**.

(53) In some embodiments, the piston of the setting tool may be configured with a plurality of axial ports to direct the exhaust parallel to the inner surface of the housing. Turning now to FIGS. **13** and **14**, another embodiment of a setting tool **300** is shown. Setting tool **300** includes features in common with the setting tool **100** shown in FIGS. **3-7**, and shared features are labeled similarly. In this exemplary embodiment, setting tool **300** generally includes a piston **302**, housing **108**, housing adapter **148**, and an exhaust diffuser **320**. Piston **302** of setting tool **300** has an annular seal assembly **310** comprising a pair of annular seals disposed on an outer surface **304**. Piston **302** additionally includes a radially outer surface **306** located downhole of the annular seal assembly **310**. Additionally, piston **302** includes a central bore ore passage **308** defined by an inner surface **312**, and a secondary bore or passage **314** extending centrally into piston **302** from central bore **308**.

(54) In this exemplary embodiment, piston **302** includes one or more circumferentially spaced exhaust ports **315** each comprising a radial port **316** and an axial port **318**. The radial port **316** of a given exhaust port **315** extends from the outer surface **304** of piston **302** to the secondary bore **314**

of piston **302**. The axial port **318** of a given exhaust port **315** extends from the end face **322** to intersect with the radial ports **316** of the piston **302**. A plug **324** may be sealingly engaged with the radial ports **316**; however, it may be understood that the configuration of exhaust ports **315** may vary in other embodiments. In this configuration, exhaust ports **315** direct a radial flow of combustion products from the secondary bore **314** to transition to an axial flow of fluid within the axial port **318**.

(55) As previously described, the housing **108** has a cylinder shape and is slidably disposed on the piston **104** with an outer surface **144** and an inner surface **146**. Housing **108** includes a housing adapter **148** releasably coupled with the housing **108**. In this exemplary embodiment, housing adapter **148** includes a seal assembly **158** in sealing engagement with outer surface **306** of the piston **302**. Additionally, piston **302** includes an annular seal assembly **310** in sealing engagement with the inner surface **146** of the housing **108** and the seal assembly **158** in sealing engagement with the outer surface **306** of the piston **302** form an annular expansion chamber **328** that is fluidically connected to the central bore **308** via the plurality of exhaust ports **330**.

(56) In this exemplary embodiment, exhaust diffuser **320** comprises a flow plug that slidingly engages the central bore **308** of piston **302** and thus may also be referred to herein as flow plug **320**. In this exemplary embodiment, Flow plug **320** is generally cylindrical in shape and includes an outer surface **332**, an uphole face **334**, a downhole face **336**, and an annular groove **338** located between the uphole face **334** and downhole face **336**. Particularly, flow plug **320** has one or more grooves **338** formed in the outer surface **332** with a front surface **340**, a back surface **342**, and a bottom surface **344**. The grooves **338** have a rectangular cross-section in this exemplary embodiment with the front surface **340** parallel to the back surface **342**. It may be understood that the geometry of grooves **338** may vary in other embodiments. For example, the groove **338** can be other shapes in the cross-section; e.g., V-shaped, U-shaped, or with curved front surface **340** and curved back surface **342**.

(57) As shown particularly in FIG. **14**, flow plug **320** includes one or more circumferentially spaced first or upstream flow ports **360** extending from the uphole face **334** to the front surface **340** of the groove **338**. Additionally, flow plug **320** includes one or more circumferentially spaced second or downstream flow ports **362** extending from the downhole face **336** to the back surface **342** of the groove **338**. Downstream flow ports **362** are axially spaced and downstream of the combustion products relative to the upstream flow ports **360** of flow plug **320**. Additionally, each of the upstream flow ports **360** in are rotated out of plane or circumferentially spaced from each of the downstream flow ports **362**. For example, a first upstream flow port **360** can be located at 90° while a first downstream flow port **362** can be located at 105° and so on and so forth such that none of the upstream flow ports **360** circumferentially align with any of the downstream flow ports **362**. In some embodiments, each downstream flow port **362** is circumferentially spaced by a predefined angular offset from at least one of the upstream flow ports **360**. The angular offset may range approximately between 5° and 30° in some embodiments; however, it may be understood that the angular offset and spacing of flow ports **360** and **362** may vary.

(58) In operation, combustion products flow from the central bore **308** to the expansion chamber **328** as the setting tool **300** strokes to actuate the auxiliary tool **44**. The combustion products flow through the flow ports **360**, into the groove **338**, and through the flow ports **362**. A portion of the flow of combustion products passes between the outer surface **332** of the flow plug **320** and the inner surface **312** of the central bore **308** of piston **302**. The restriction to fluid flow through the flow plug **320** may be adjusted depending on the number and geometry of the flow ports **360** and the flow ports **362**. The flow of combustion products pass through the secondary bore **314** to enter the exhaust ports **330**. The flow of combustion products transitions from radial flow direction within the radial ports **316** to an axial flow direction within the axial ports **318**. The flow of combustion products may be restricted through the plurality of exhaust ports **330** depending on the number and geometry of the exhaust ports **330**. The flow exiting the exhaust ports **330** is parallel to

the inner surface **146** of the housing **108**.

(59) Referring now to FIG. **15**, an embodiment of a method **400** for redressing a setting tool for actuating a plug in a subterranean wellbore is shown. Beginning at block **402**, method **400** comprises recovering the setting tool from the wellbore, the setting tool comprising a housing with a throughbore therein, a piston arranged for axial movement within the throughbore, a combustible element, and an exhaust diffuser configured to redirect a flow of combustion products generated in response to ignition of the combustible element. In some embodiments, block **402** comprises recovering one of the setting tools **100** and **300** described above from the wellbore **14**.

(60) At block **404**, method **400** comprises removing the piston from the throughbore of the housing. In some embodiments, block **404** comprises removing the piston **104** from the throughbore of the housing **108** of setting tool **100**. At block **406**, method **400** comprises removing the exhaust diffuser from the housing. In some embodiments, block **406** comprises removing one of the exhaust diffusers **184**, **230**, **250**, and **32** described herein from the housing **108**. At block **408**, method **400** comprises cleaning the piston and the housing. In some embodiments, block **408** comprises removing foreign materials or debris that have accumulated onto the piston and the housing of the setting tool (e.g., onto the piston **104** and the housing **108** of setting tool **100**). The removal of debris from the piston and housing may include buffing, scraping, heating, washing, treating via one or more chemicals, and other techniques for cleaning the piston and the housing. Additionally, consumable parts such as elastomeric seals, fasteners, etc., of the piston and the housing may be replaced to redress the setting tool.

(61) At block **410**, method **400** comprises reinstalling the piston into the housing of the setting tool with at least one of the exhaust diffuser and a replacement exhaust diffuser for subsequent use of the setting tool in a wellbore. In some embodiments, block **410** comprises reinstalling the piston **104** into the housing **108** of setting tool **100** with at least one of the exhaust diffusers **184**, **230**, **250**, and **32** described herein, where the exhaust diffuser may be the previously used exhaust diffuser or a new, unused exhaust diffuser. For example, in some embodiments, the previously used exhaust diffuser may be cleaned prior to being reinstalled in the housing. In this manner, foreign materials or debris may be removed from the exhaust diffuser in a manner similar to the cleaning of the piston and the housing briefly addressed above.

(62) While exemplary embodiments have been shown and described, modifications thereof can be made by one skilled in the art without departing from the scope or teachings herein. The embodiments described herein are exemplary only and are not limiting. Many variations and modifications of the systems, apparatus, and processes described herein are possible and are within the scope of the disclosure presented herein. For example, the relative dimensions of various parts, the materials from which the various parts are made, and other parameters can be varied. Accordingly, the scope of protection is not limited to the embodiments described herein, but is only limited by the claims that follow, the scope of which shall include all equivalents of the subject matter of the claims. Unless expressly stated otherwise, the steps in a method claim may be performed in any order. The recitation of identifiers such as (a), (b), (c) or (1), (2), (3) before steps in a method claim are not intended to and do not specify a particular order to the steps, but rather are used to simplify subsequent reference to such steps.

Claims

1. A method for redressing a setting tool for actuating a plug in a subterranean wellbore, the method comprising: (a) recovering the setting tool from the wellbore, the setting tool comprising a housing with a throughbore therein, a piston arranged for axial movement within the throughbore, a combustible element, and an exhaust diffuser configured to redirect a flow of combustion products generated in response to ignition of the combustible element, wherein the setting tool comprises an initial configuration with the piston located in a first axial position in the housing, and a set

configuration with the piston located in a second axial position in the housing spaced from the first axial position, the piston being displaced from the first axial position to the second axial position by the flow combustion products, and wherein fluid communication is permitted through the exhaust diffuser in both the initial configuration and the set configuration; (b) removing the piston from the throughbore of the housing; (c) removing the exhaust diffuser from the housing; (d) cleaning the piston and the housing; and (e) reinstalling the piston into the housing of the setting tool with at least one of the exhaust diffuser and a replacement exhaust diffuser for subsequent use of the setting tool in a wellbore whereby at least one of the exhaust diffuser and the replacement exhaust diffuser is positioned in an annular expansion chamber formed radially between an outer surface of the piston and an inner surface of the housing.
2. The method according to claim 1, wherein (e) comprises orienting at least a portion of a baffle face of at least one of the exhaust diffuser and the replacement exhaust diffuser at an angle that is between 60° and 120° relative to a longitudinal axis of an exhaust port of the piston.
3. The method according to claim 1, wherein (e) comprises positioning at least one of the exhaust diffuser and the replacement exhaust diffuser in an internal firing chamber formed within the piston.
4. The method according to claim 1, wherein (e) comprises releasably coupling at least one of the exhaust diffuser and the replacement exhaust diffuser to the piston.
5. The method according to claim 1, wherein (e) comprises releasably coupling at least one of the exhaust diffuser and the replacement exhaust diffuser to the housing.
6. The method according to claim 1, wherein (e) comprises reinstalling the piston into the housing of the setting tool with the replacement exhaust diffuser.
7. A method for redressing a setting tool for actuating a plug in a subterranean wellbore, the method comprising: (a) recovering the setting tool from the wellbore, the setting tool comprising a housing with a throughbore therein, a piston arranged for axial movement within the throughbore, a combustible element, and an exhaust diffuser configured to redirect a flow of combustion products generated in response to ignition of the combustible element, wherein the setting tool comprises an initial configuration with the piston located in a first axial position in the housing, and a set configuration with the piston located in a second axial position in the housing spaced from the first axial position, the piston being displaced from the first axial position to the second axial position by the flow combustion products, and wherein fluid communication is permitted through the exhaust diffuser in both the initial configuration and the set configuration; (b) removing the piston from the throughbore of the housing; (c) removing the exhaust diffuser from the housing; (d) cleaning the exhaust diffuser, the piston, and the housing; and (e) comprises reinstalling the piston into the housing of the setting tool with the cleaned exhaust diffuser for subsequent use of the setting tool in a wellbore.